Final Report

Prediction of Airline Customer Churn

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Executive Summary

This project addresses this critical business challenge by developing a data-driven predictive system to identify high-value customers at risk of churn before they disengage. Utilizing a comprehensive dataset from a Chinese airline's frequent-flyer program, we constructed a robust binary churn indicator (is_churn) by analyzing customer activity patterns over time. Through rigorous feature engineering and selection, we identified and retained eight core business-relevant features—such as weighted flight mileage, historical flight count, and average discount—while meticulously avoiding data leakage.

Multiple machine learning models, including Logistic Regression, Random Forest, SVM, LightGBM, and XGBoost, were developed and evaluated alongside a deep learning Multilayer Perceptron (MLP). While the MLP demonstrated superior predictive performance, the XGBoost model was selected as the optimal deployment model due to its excellent balance of high performance, strong interpretability, and operational efficiency. The chosen XGBoost model provides highly actionable insights. Feature importance analysis reveals that Average Discount, Weighted Flight Mileage, and Customer Age are the most critical factors influencing churn risk. Based on these insights, targeted retention strategies are recommended, including personalized discount incentives for price-sensitive customers, exclusive loyalty benefits for historically high-value members, and age-group-specific engagement campaigns.

In conclusion, this project successfully delivers a highly accurate High-Value Customer Churn Early Warning System. It empowers the airline to shift its customer relationship management paradigm from reactive to proactive, enabling precise, timely, and cost-effective retention interventions. This transformation directly safeguards revenue by converting potential churn into retained loyalty, thereby enhancing competitive advantage and profitability.

1.Introduction

1.1 Industry Background and Strategic Significance

Aviation industry is very competitive with customer loyalty being the key to profitability in the airline. With the growing competitive pressure within the market, airlines have to be more precise in determining customers at risk of churn and have to apply efficient retention strategy. The old method of churn analysis is usually done after the customers are gone and in such a situation, the reacquisition costs are expensive, and the success of reacquisition is low. Consequently, having a model that will be able to foretell customer risk will have great commercial importance to airlines.

1.2 Core Project Objectives

The proposed project will use the historical data of customer behavior in the airline to perform customer behavior analysis and churn prediction modelling.

First, we perform some basic preprocessing on the airline passenger dataset: we remove the features that could may cause data leakage and invalid ones. At this point, we filter out a set of core features that are obviously important to the business. This will enable us to come up with customers who are at high risk of being churned.

Next, we build several churn predictive models and evaluate their performance. We end up choosing the best performing model as the main deployment model depending on the main metrics, including churn recall rate and AUC.

By using this prediction model, the airline could change its customer churn management model to be proactive remediation instead of reactive intervention. Based on the churn alerts and feature insights provided by the model, one can design specific recovery strategies on the high-value inactive customers to help protect the company revenues and profits.

1.3 Problem Statement

Conventional methods utilize experience-based judgements and straightforward rules which give rise to 3 significant problems: slowness in preemptive warnings, incorrect target recognition, and inefficient strategy performance.

The existing customer attrition detection mainly uses a mere threshold assessment of historical flight information. When a customer is officially declared as lost, they usually have nothing scheduled in terms of flights more than six months ago, which makes recovery operations very unlikely to succeed and thus very expensive. The old techniques do not discriminate among the various categories of dormant customers: old active but currently inactive but high-value

customers (key retention objectives), dormant customers who never flew, and customers who have temporarily stopped flying because of temporary causes. Without support data, retention strategies are also not able to develop personalized intervention plan based on historical value, behavioral characteristics, and churn risk level leading to low overall retention effectiveness.

2. Dataset Description

2.1 Dataset Overview

The "Airline Customer Dataset" is a collection of flight records and membership profiles , which is gathered from a Chinese airline's frequent-flyer program.

The information of this dataset is accessible to the public on the Heywhale platform.

(https://www.heywhale.com/mw/dataset/6007f3e47ed5ab0015f250a6/file).

We can use this dataset to segment customers and analyze loyalty.

2.2 Dataset Structure

There are 62,988 rows and 44 columns in this dataset, including member information, flight history, and point accumulation records. This dataset records data from 2004 to 2015, each row of it corresponds a passenger. These attributes can be categorized into several groups, such as personal information, flight behavior, and loyalty program performance.

2.3 Feature Description

The dataset contains 44 variables in total, which can be classified into six major categories as below.

Category	Number of Features	Example Features	Data Type
Identification	1	MEMBER_NO	Numerical
Member Information	5	GENDER, AGE, WORK_CITY, WORK_PROVINCE, WORK_COUNTRY	Categorical/ Numerical
Date-related	4	FFP_DATE, FIRST_FLIGHT_DATE, LAST_FLIGHT_DATE, LOAD_TIME	Date
Flight Behavior	10	FLIGHT_COUNT, SEG_KM_SUM, AVG_INTERVAL, MAX_INTERVAL, BEGIN_TO_FIRST	Numerical

Loyalty & Points	15	BP_SUM, EP_SUM_YR_1, EP_SUM_YR_2, ADD_POINTS_SUM_YR_1, EXCHANGE_COUNT	Numerical
Ratio & Derived Metrics	9	avg_discount, Ratio_LY1_BPS, Ratio_L1Y_BPS, Ratio_P1Y_Flight_Count	Numerical

2.4 Estimation of the Target Variable

In this research, a binary target variable is created, which is called is_churn, and it is evaluated with respect to the flight and accrued miles activity of the users in the airline system in the last two years. This variable will be used to describe the group of customers that used to be active but is currently dormant by determining the relationship between the current dormant status and the historical active status. It is used to determine dormant risk and churn groups.

2.4.1 Foundational Elements of the Target Variable

There are a number of important concepts that should be explained when building the y variable.

- L1Y_*: Variables with L1Y in the prefix denote the activity of a customer during the past 1 year (L1Y_Flight_Count and L1Y_Point_Sum), the number of flights made and the cumulative loyalty points of the customer respectively in the last year.
- P1Y_*: P1Y_ Flight Count is a variable with the prefix P1Y indicating that the variable describes the activity of a customer in the last 1 year (P1Y_Flight Count). It records the past behavior and historical flight.
- silent_now: This is an intermediate variable that shows whether a customer is currently inactive or not. In particular silent_now will be True in case the customer has not been flying or earning points during the previous year, and

False otherwise.

- active_prev: The other intermediate variable to capture the historical activity of the customer is active_prev which is defined as True when the customer recorded any flight or point activity in the past year or before.
- y / is_churn: The last target variable that shows silent risk. A customer is considered to be is_churn = 1 when the user is an active customer in the past (active_prev = True), and now silent (silent_now = true), is considered a silent risk customer. The rest of the customers are identified as is_churn = 0, including the continuously active customers, the new customers or the customers who do not have enough history.

2.4.2 Calculation Steps

The process of calculation of is_churn is based on three fundamental steps.

The first step is to calculate silent_now, which is the current state of silence. A customer is deemed as being currently silent only when he or she exhibits no core activity during the last year: he or she made no flights, no base points earned, and no further points accrued. In case any of these activities are used to take place, the customer is not identified as silent (silent_now = False).

The second step is to compute active_prev, the historical active status. In this case, we will verify whether the customer has been active during the previous year prior to the last, or even the previous year. We apply several backup indicators, including the number of flights or the earnings of points of the previous year to prevent potential misclassification because of any gaps in the data. When any of these signals are yes that there was activity in the past, active_pre is established to True. Otherwise, the customer is treated as one who has never been active.

The third step is the calculation of the ultimate target variable y by silent_now and active_prev. The essence will be to determine those customers who have become historically active but are now silent. The flagging a customer as a churn risk (y = 1) is done by meeting the two conditions simultaneously:

1. They were active previously (active_prev = True)
2. They are now silent (silent_now = True).

In case of non-compliance with either of the conditions, the value of y = 0 is given, that is, the customer is not in the at-risk churn group.

silent_now	active_prev	y	Type
True	True	1	Silent Risk Customer (Key Focus Group)
False	True	0	Continuously Active Customer
True	False	0	New Customer or Never Active Customer
False	False	0	Abnormal or Noise Type (Rare Occurrence)

3. Feature Selection

3.1 Features Removed to Avoid Data Leakage

During feature selection, we removed all variables that were involved in constructing the is_churn label, such as the variables include the L1Y_* variables, Ration_* variables, and long-term cumulative variables. We avoid using these variables that would cause data leakage because they show information from the prediction period.

3.2 Features Not Suitable For Modeling

We also removed features that were not helpful for prediction, including ID numbers, original timestamps and geographical attributes. These variables provide little useful information and probably introduce noise into the

model.

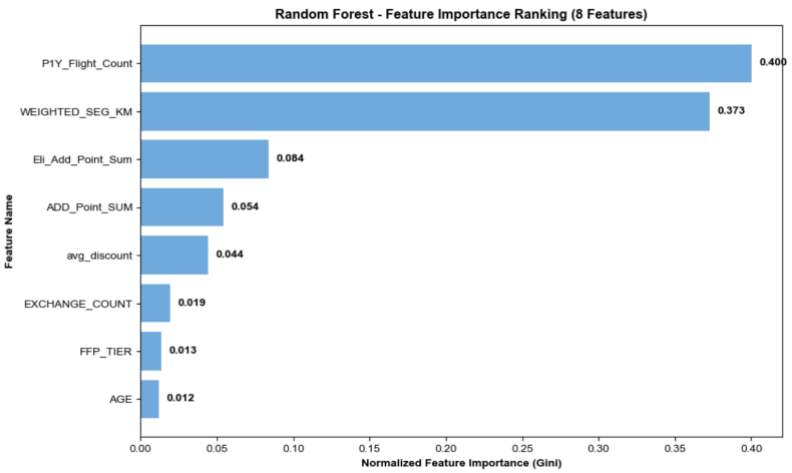
3.3 Feature Selection

After removing certain features, there exist 8 features. These features describe customer value, activity level, price sensitivity, and engagement. However, we still need to measure the feature importance to select the final selected feature.

Feature	Meaning
FFP_TIER	Membership level
AGE	Customer age
WEIGHTED_SEG_KM	Weighted flight mileage
EXCHANGE_COUNT	Number of points redemptions
avg_discount	Average ticket discount
P1Y_Flight_Count	Flight count in the previous year
ADD_Point_SUM	Partner, promotional, or external points
Eli_Add_Point_Sum	Non-flight points

3.3.1 Feature importance Evaluation Using RandomForest

In order to consider the nonlinear relationship of features, we adopted the RandomForestClassifier model to measure the feature importance. The results are as follows:



The combined importance of WEIGHTED_SEG_KM and P1Y_Flight_Count exceeds 60%, which can also verify that

flight mileage and flight frequency are the key variables for prediction. The third important feature is Eli_Add_Point_Sum, it has a relatively high contribution in the tree model.

3.3.2 Comparison of The Different Feature Set

Based on the results of the above feature selection method, the following conclusions can be drawn:

user behavior characteristics: WEIGHTED_SEG_KM and P1Y_Flight_Count rank high in RandomForest importance.

Important supplementary features: Eli_Add_Point_Sum and ADD_Point_SUM contribute significantly.

We built two versions of the XGBoost model to compare the feature selection effects:

Model A: Only use the following four core features: WEIGHTED_SEG_KM, P1Y_Flight_Count, Eli_Add_Point_Sum, ADD_Point_SUM

Model B: Use all 8 candidate features.

The results are as follows:

Performance	**Model A**	**Model B**
AUC	0.9302	0.9411
Accuracy	0.8430	0.8489
Recall	0.8835	0.8826
Precision	0.5020	0.5257
F1-Score	0.6402	0.6589

It can be seen that the model B containing all 8 features has a better performance than the model A using only 4 features in various indicators such as AUC, Accuracy rate, Precision rate and F1-Score.

3.3.3 Conclusion

The results show that although using only four core features can achieve a good prediction effect , But the overall ability

of the model has been improved after introducing supplementary variables such as integration and grade. Therefore, we adopt eight features in the following prediction.

4. Dataset Split

To ensure stable model training and reliable evaluation, we split the dataset into training, validation, and test sets using an 8:1:1 ratio.

Dataset	Proportion	Core Usage
Training Set	80%	Used for feature standardization fitting, grid search cross-validation, and model training; only participates in the core process of parameter tuning
Validation Set	10%	Used to evaluate the performance of the optimal model selected by grid search and assist in judging overfitting (does not participate in training)
Test Set	10%	Serves as an independent dataset, only used for verifying the generalization ability of the final model; does not participate in any tuning/training processes

5. Model Descriptions

5.1 Machine Learning Models

5.1.1 Logit Regression

Reasons for Model Selection

Logistic Regression is adopted as the baseline model for customer churn prediction in this project, with core selection criteria balancing business feasibility and technical adaptability as follows:

Strong business interpretability: Logistic Regression outputs the probability of customer churn (ranging from 0 to 1), and each feature corresponds to a clear standardized coefficient. It can directly quantify the direction (positive coefficient = promoting churn, negative coefficient = inhibiting churn) and intensity (larger absolute value of the coefficient indicates stronger impact) of the feature's influence on churn risk, providing interpretable decision-making basis for formulating hierarchical retention strategies.

Adaptability in training efficiency and robustness: As a linear model, Logistic Regression can converge quickly with the 100,000-level sample size in this project and is tolerant to minor outliers in data (such as extreme consumption records).

Value as a baseline model: The performance of Logistic Regression can serve as a benchmark for subsequent complex models (such as XGBoost and LightGBM). By comparing core metrics like AUC and recall rate, the performance improvement of complex models can be clearly quantified, verifying the effectiveness of model iteration.

Handling Class Imbalance

The customer churn rate in this project is approximately 15.8% (proportion of positive samples), which is a typical binary classification scenario with class imbalance. Logistic Regression achieves class balance at the loss level through the `class_weight` parameter, with the core logic as follows:

Core method: The `class_weight='balanced'` mode is adopted, where the model automatically calculates weights based on the ratio of positive and negative samples in the training set (weight formula: $w_j = n_{samples} / \left(n_{classes} \times n_{samples_j} \right)$. In this project, by amplifying the misclassification penalty for positive samples, the model is forced to pay attention to minority

class features during training, avoiding the problem of "fitting dominated by the majority class".

Advantage: This method only adjusts the loss weight through parameter configuration, without the need for oversampling (such as SMOTE) or undersampling of the original data. It retains the information of all real business data, avoids the noise risk introduced by synthetic samples, and improves the generalization ability of the model.

Parameter Tuning

Core Tuning Parameters and Grid Search Design

Based on the core characteristics of Logistic Regression, a streamlined and efficient parameter grid is designed, with only "regularization strength" and "solver" tuned, while other parameters are fixed (to ensure focused tuning):

Tuning Parameter	Candidate Range	Parameter Function
C	[0.01, 0.1, 1, 10, 100]	Reciprocal of regularization strength (smaller C means stronger L2 regularization to avoid overfitting; larger C means stronger fitting ability)
solver	['liblinear', 'saga']	Optimization solver: liblinear is suitable for small-sample binary classification; saga supports elastic-net regularization to improve adaptability
Fixed Parameters	-	class_weight='balanced', max_iter=1000, random_state=42

The optimal parameter combination selected through grid search is as follows:

- C=100 (regularization strength is adapted to this dataset to avoid overfitting);
- solver='liblinear' (fastest convergence speed and optimal stability in binary classification scenarios);

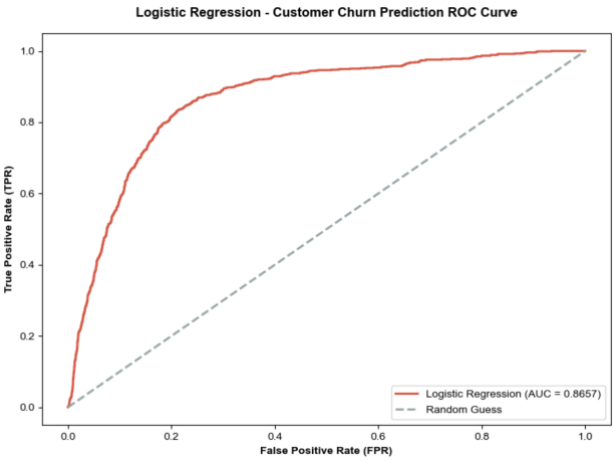
- Other parameters: class_weight='balanced', max_iter=1000.

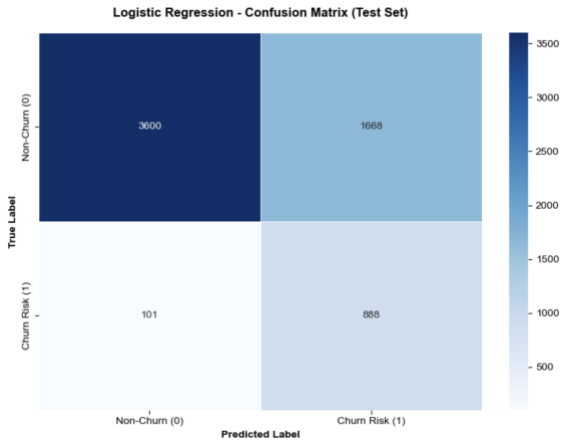
Final Verification: Evaluate the generalization ability of the optimal model on the test set and output the final metrics.

Optimal Parameter Results

Model Result Output

ROC Curve and AUC Value& Confusion Matrix：





- Churn Recall: 0.8979 (covers 89.8% of real churners), AUC: 0.8657 (≥0.85 = Good for linear models), Churn Precision: 0.3474 (reduces false positive waste), F1 Score: 0.501 (trade-off between recall and precision)

5.1.2 Random Forest

Reasons for Model Selection

Random Forest is a tree-based model based on ensemble learning. The core reasons for selecting this model are as follows:

Strong adaptability to structured data (such as aviation customer characteristic data), enabling direct modeling without feature standardization, thus reducing data preprocessing costs;

Natural resistance to overfitting (through random sampling of multiple decision trees and random feature selection),

with better stability than a single decision tree on small and medium-sized datasets;

Capable of outputting feature importance, facilitating the analysis of core factors affecting customer churn and meeting the requirement of business interpretability;

High training efficiency and low parameter tuning cost, making it suitable as a baseline model.

Handling Class Imbalance

To address the imbalance issue of "low proportion of churned samples" in customer churn data, the `class_weight='balanced'` parameter is adopted:

This parameter automatically adjusts weights according to the class distribution of samples, assigning higher weights to the minority class (churned customers) and lower weights to the majority class (non-churned customers);

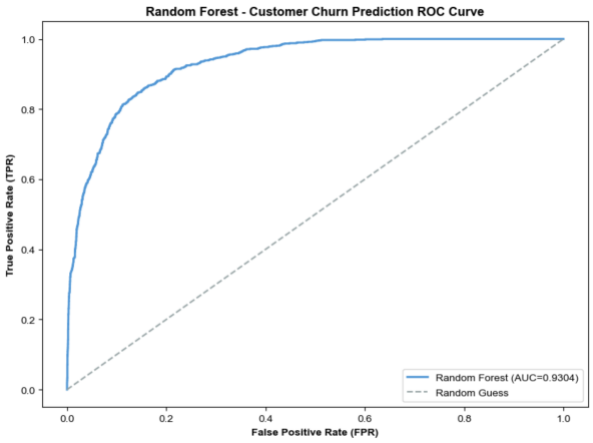
Core logic: Weight = Total number of samples / (Number of classes × Number of samples in the class), ensuring the model does not favor the majority class during training.

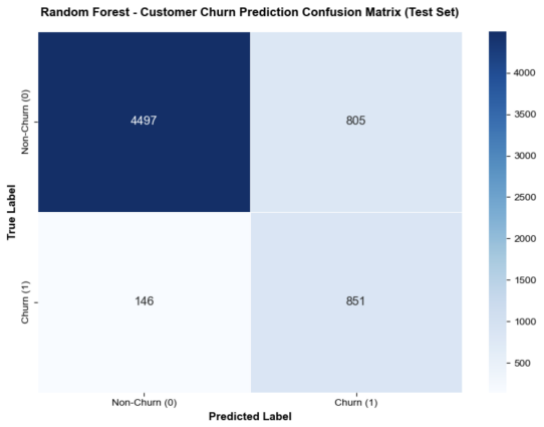
Parameter Tuning

- Parameter tuning range: Core tuning parameters include the number of decision trees (`n_estimators: [50, 100, 150]`), maximum tree depth (`max_depth: [6, 8, 10]`), and minimum number of samples required to split a node (`min_samples_split: [10, 20]`);
- Optimal parameter: {'class_weight': 'balanced', 'max_depth': 10, 'min_samples_split': 20, 'n_estimators': 50, 'random_state': 42}
- Final evaluation: The final model metrics are output on the independent test set to avoid overfitting.

Optimal Parameter Results

Model Result Output





- AUC: 0.9304, Churn Recall: 0.8536 (Covers 85.4% of actual churners), Churn Precision: 0.5139, F1 Score: 0.6415, Accuracy: 0.849

5.1.3 SVM

Reasons for Model Selection

SVM is a classification model based on maximum margin. The core reasons for selecting this model are as follows:

- Suitable for classification tasks in high-dimensional feature spaces, capable of effectively capturing the nonlinear relationships between customer features and churn behavior;
- Based on the "maximum margin" principle, it has good generalization ability on small-sample datasets and is not prone to overfitting;
- Can map linearly inseparable data in low-dimensional space to high-dimensional space through kernel functions (such as RBF kernel) to solve nonlinear classification problems;
- As a classic classification algorithm, it can be compared with tree-based models to verify the rationality of algorithm selection.

Handling Class Imbalance

1. Weight adjustment: Use the `class_weight='balanced'` parameter to inversely adjust the penalty weights according to the class sample proportion, alleviating the problem of the minority class (churned customers) being ignored;

2. Feature standardization: SVM is sensitive to feature scales. All features are standardized to a mean of 0 and a variance of 1 through `StandardScaler` to avoid features with large numerical range differences (such as WEIGHTED_SEG_KM) dominating model training;

3. Pipeline integration: Integrate "standardization + modeling" through `Pipeline` to ensure the separation of standardization parameters between the training set and the test set (the test set uses the mean/variance of the training set) and avoid data leakage.

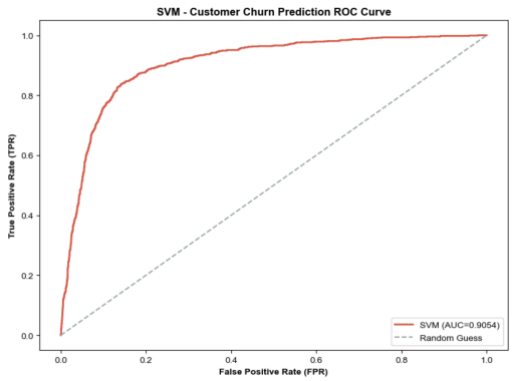
Parameter Tuning

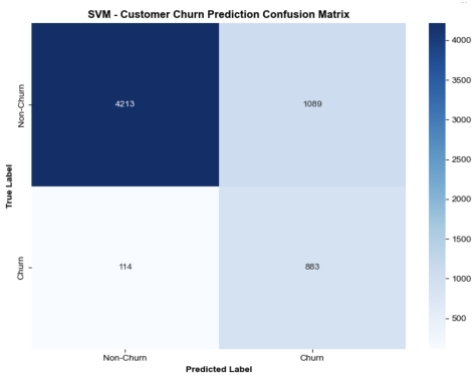
- Parameter tuning range: Core tuning parameters include penalty coefficient (C: [0.1, 1, 10]), kernel function coefficient (gamma: ['scale', 'auto']), and kernel function type (kernel: ['rbf', 'linear'])

- Optimal parameters: {'svm__C': 10, 'svm__class_weight': 'balanced', 'svm__gamma': 'scale', 'svm__kernel': 'rbf'}

- **Optimal Parameter Results**

Model Result Output





- AUC: 0.9054, Churn Recall: 0.8857 (covers 88.6% of actual churn customers), Churn Precision: 0.4478, F1 Score: 0.5948 Accuracy: 0.809

5.1.4 LightGBM

Reasons for Model Selection

LightGBM is a gradient boosting tree based on histograms. The core reasons for selecting this model are as follows:

Adopts "histogram algorithm + one-side gradient sampling", with faster training speed and lower memory usage than XGBoost, making it suitable for large-scale data; Supports direct processing of categorical features and has better adaptability to imbalanced data; The core parameter num_leaves can flexibly control model complexity, balancing fitting ability and generalization ability; Also supports feature importance output and has higher training efficiency than XGBoost, making it suitable for rapid iterative tuning.

Handling Class Imbalance

1. Weight adjustment: Set scale_pos_weight = Number of negative samples / Number of positive samples, consistent with XGBoost logic, to balance the loss weights of positive and negative samples;

2. Parameter optimization: Reduce the minimum number of samples per leaf node through min_child_samples to ensure that minority class samples can form effective splits;

Parameter Tuning

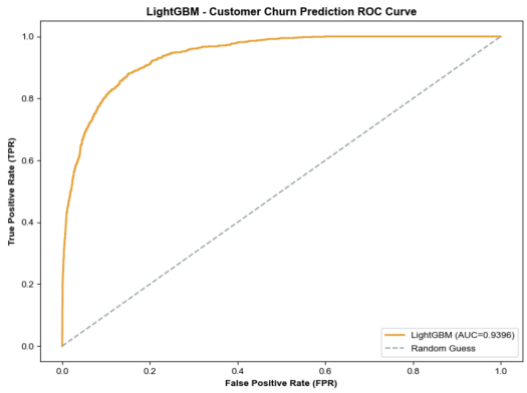
- Parameter tuning range: Core tuning parameters include the number of trees (n_estimators: [50, 100]), tree depth (max_depth: [5, 7]), learning rate (learning_rate: [0.1, 0.2]), number of leaf nodes (num_leaves: [63, 127]), and

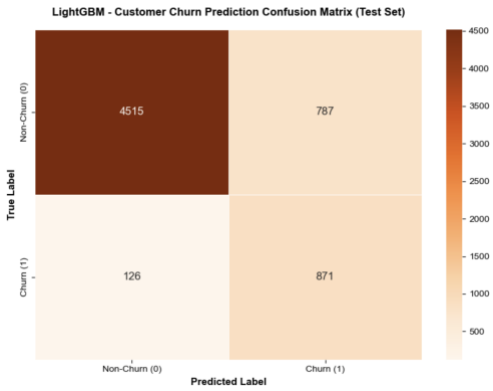
sampling ratio (subsample: [0.8, 1.0])

- Best parameters: {'learning_rate': 0.2, 'max_depth': 5, 'n_estimators': 100, 'num_leaves': 63, 'scale_pos_weight': 5.32, 'subsample': 0.8}

Optimal Parameter Results

Model Result Output





- AUC: 0.9396, Churn Recall: 0.8736 (Covers 87.4% of actual churners), Churn Precision: 0.5253, F1 Score: 0.6561, Accuracy: 0.8551

5.1.5 XGBoost

Reasons for Model Selection

XGBoost is an optimized version of Gradient Boosting Decision Tree (GBDT). The core reasons for selecting this model are as follows:

- Adopts regularization terms (L1/L2) and column sampling, with better anti-overfitting ability than traditional GBDT and superior performance in structured data classification tasks;

- Supports automatic handling of missing values, enabling adaptation to customer data containing missing values

without additional filling logic;

- High training efficiency (parallel processing) and can output feature importance, balancing performance and interpretability;
- Provides the scale_pos_weight parameter for imbalanced data, accurately adapting to the churn prediction scenario.

Handling Class Imbalance

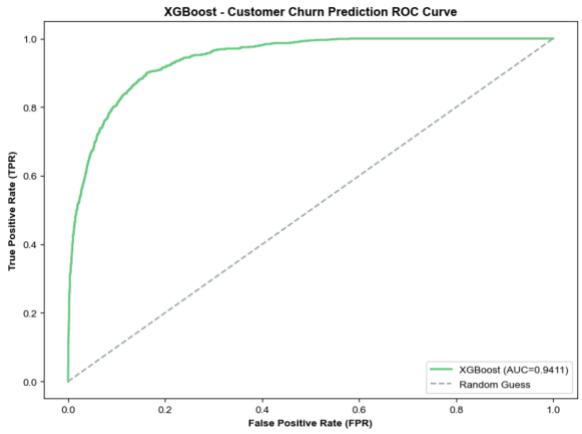
1. Weight adjustment: Set scale_pos_weight = Number of negative samples / Number of positive samples (i.e., Number of non-churned customers / Number of churned customers) to directly adjust the weights of positive and negative samples in the loss function, making the model pay more attention to the minority class;
2. Evaluation metric adaptation: Use ROC-AUC instead of accuracy as the core evaluation metric to avoid the problem of "high accuracy but low recall rate" caused by data imbalance;
3. Parameter constraints: Limit the tree depth through max_depth to prevent the model from overfitting to the features of the majority class.

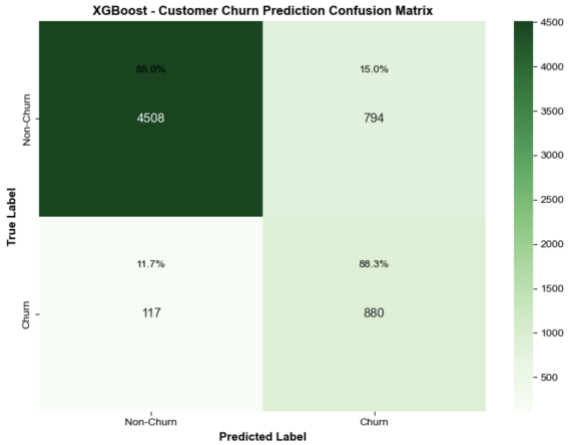
Parameter Tuning

- Parameter tuning range: Core tuning parameters include the number of trees (n_estimators: [50, 100, 150]), tree depth (max_depth: [3, 5, 7]), learning rate (learning_rate: [0.01, 0.1, 0.2]), and sampling ratio (subsample: [0.8, 1.0])
- Optimal parameter: {'learning_rate': 0.2, 'max_depth': 5, 'n_estimators': 100, 'scale_pos_weight': 5.32, 'subsample':

1.0}

Model Result Output





- AUC: 0.9411, Churn Recall: 0.8826 (covers 88.3% of actual churn customers), Churn Precision: 0.5257, F1 Score: 0.6589, Accuracy: 0.8554

5.2 Deep Learning Model

The Multilayer Perceptron (MLP) is introduced as a deep learning–based benchmark for the churn prediction task. Unlike tree-based models such as XGBoost and Random Forest, the MLP learns a nonlinear mapping from input features to churn probability through stacked fully connected layers and gradient-based optimization, allowing it to capture complex interactions among customer profile, flight behavior and loyalty variables.

5.2.1 Input Features and Preprocessing

To ensure a fair comparison and avoid data leakage, the MLP uses the same leakage-safe feature set as XGBoost: only variables observable at prediction time, excluding all L1Y_* and ratio variables that directly encode information from

the prediction period. The final feature vector contains nine original numerical variables describing long-term value and behavior (FFP_TIER, AGE, WEIGHTED_SEG_KM, LAST_TO_END, EXCHANGE_COUNT, avg_discount,P1Y_Flight_Count,ADD_Point_SUM, Eli_Add_Point_Sum). Several are highly skewed with many zeros and a few extremely large values, so two log-transformed variables, log1p_ADD_Point_SUM and log1p_WEIGHTED_SEG_KM, are added to compress the long tails.

All 11 numeric features are then standardized by RobustScaler, which scales by the median and interquartile range and is therefore more robust to outliers. This step is important for stable training of gradient-based models such as the MLP.

5.2.2 Network Architecture and Training Setup

The proposed MLP is a four-layer fully connected network:

- Input layer: dimension equal to the preprocessed feature vector.
- Hidden Layer 1: 512 neurons, ReLU activation.
- Hidden Layer 2: 256 neurons, ReLU activation.
- Hidden Layer 3: 128 neurons, ReLU activation.
- Output layer: 1 neuron, Sigmoid activation to produce churn probability y^∈[0,1]\hat{y} \in [0,1]y^ ∈[0,1].

A dropout rate of 0.5 is applied between hidden layers to mitigate overfitting. The network is trained with the Adam optimizer (learning rate 10−310^{-3}10−3) and binary cross-entropy loss. The data are split into training/validation/test sets with an 80%/10%/10% stratified partition, consistent with the other models. Early stopping based on validation performance is used to prevent overfitting.

Figure X (training and validation AUC curves) shows that both AUC values rise rapidly above 0.99 and then stay stable, with the validation curve closely tracking the training curve. This indicates fast convergence and no obvious overfitting on the validation set.

5.2.3 Results and Visualization

On the held-out test set, the MLP achieves:

- AUC: 0.9923, Recall: 0.9443, Precision: 0.9953, F1-score: 0.9691, Accuracy: 0.9905

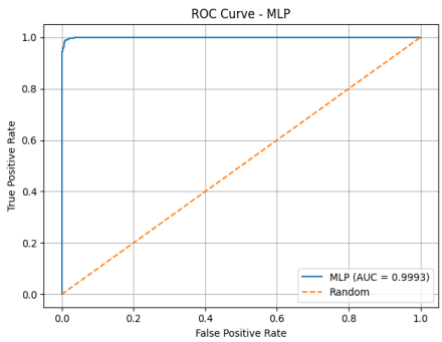


Figure 1

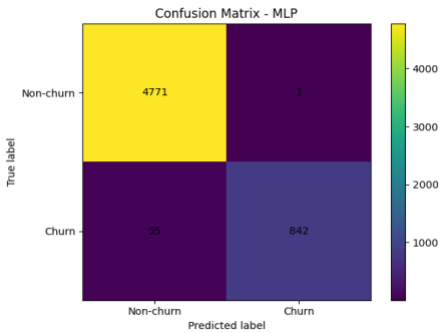


Figure 2

The ROC curve in Figure 1 is almost attached to the top-left corner, confirming near-perfect discrimination between churners and non-churners. The confusion matrix in Figure 2 further illustrates the error pattern: out of 4,772 non-churn customers, 4,771 are correctly classified and only one is incorrectly flagged as churn; among 897 churners, 842 are correctly detected while 55 are missed. This explains the combination of extremely high precision (very few false positives) and high recall (most churners correctly identified).

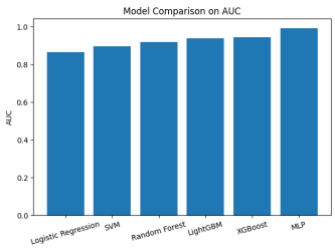


Figure 3

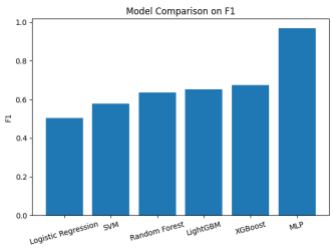


Figure 4

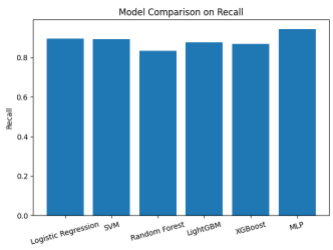


Figure 5

Figures 3-5 compare the MLP with Logistic Regression, SVM, Random Forest, LightGBM and XGBoost in terms of AUC, F1-score and recall. Across all three metrics, the MLP clearly outperforms the baselines: its AUC is substantially higher, its F1-score is much larger than even the best tree-based model, and it also attains the highest recall. In other words, the MLP recovers more churners while generating far fewer false alarms, offering the most favorable trade-off for targeted retention campaigns.

6.1 Model Selection

The following table compares the performance of all candidate models across key evaluation metrics. It can be seen that XGBoost and MLP are the top two best-performing models, and XGBoost belongs to a machine learning model while MLP belongs to a deep learning model.

Model	AUC	Recall	Precision	F-measure	Accuracy
Logistic Regression	0.8657	0.8979	0.3474	0.501	0.7216
SVM	0.9054	0.8857	0.4478	0.5948	0.809
Random Forest	0.9304	0.8536	0.5139	0.6415	0.849
LightGBM	0.9396	0.8736	0.5253	0.6561	0.8551
XGBoost	**0.9411**	**0.8826**	**0.5257**	**0.6589**	**0.8554**
MLP	**0.9923**	**0.9443**	**0.9953**	**0.9691**	**0.9905**

After conducting a comparative analysis of the advantages and disadvantages of the top two models with the best performance, the following conclusion was drawn:

Although MLP performs better than XGBoost in core indicators, given that the MLP model is more difficult to explain and understand, it does not have as good interpretability as XGBoost. It can visually present the model's prediction process and help us output feature importance charts and prediction results. Therefore, after comparing these models, we chose the XGBoost model, which performs best in terms of model performance, interpretability and cost. And determine our current stage strategy: adopt the XGBoost model as the core model of the customer churn prediction project and the future exploration direction of the MLP model project.

6.2 Business Interpretation of XGBoost's Core Metrics

The XGBoost model's performance is demonstrated through these key metrics:

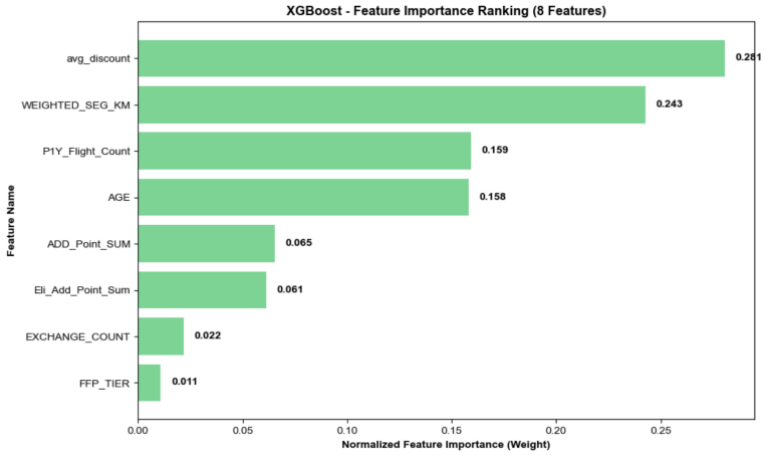
(1) **Recall (0.8826) :** This indicates that our model can capture 88 out of every 100 truly lost high-value customers, demonstrating a high recognition rate and effectively preventing the omission of important customers.

(2) **Precision (0.5257) :** The accuracy rate exceeds 52%. In every two early warnings of our prediction project, there will be one high-value customer who is truly lost. Given that the orientation of our project is to identify as many lost customers as possible, a slightly lower accuracy rate can be accepted. Retention intervention for the warned customers also has a certain positive effect on the customers who are not the target customers in the early warnings.

(3) **AUC (0.9411) :** The AUC performance is excellent. The project has a relatively outstanding ranking ability, which can help us always focus on the clients with the highest risk and value, and allocate resources efficiently.

(4) **F1 Score (0.6589) & Accuracy (0.8554**： The strong F1 score demonstrates an excellent balance between recall and precision, while the high accuracy rate of 85.54% confirms the model's overall reliability in classification tasks.

6.3 Model Insights and Strategic Recommendations



Based on the feature importance analysis chart in the above figure, the top features determined by the XGBoost model are: Average Discount, Weighted Mileage, P1Y_Flight_Count and Age. They are all key factors in identifying the loss of high-value customers. For the three most important indicators, we propose the following strategic suggestions to achieve the goals of precise marketing and improving the efficiency of resource allocation:

6.3.1 Average Discount- For price-sensitive customers

Price-sensitive customers often carefully compare prices when making a purchase to look for the most cost-effective option. We can issue coupons of different levels based on customers' historical consumption levels, such as "Exclusive membership coupons" and "time-limited discount coupons", to stimulate customers' repurchase and increase their purchase frequency.

6.3.2 Weighted Mileage- For historical high-value customers

Customers with a higher weighted mileage but a recent decline in activity should be given special attention. Compared with new users, the loss of these customers causes the greatest damage to the company. Their silence means a break in loyalty. The company needs to reach out through multiple channels to enhance customer satisfaction, offer exclusive

membership channels, free cabin upgrades, VIP lounges and other privileged services to retain customers and rebuild their loyalty.

6.3.3 P1Y_Flight_Count-For customers with different flight counts

For customers with different flight frequencies, offer exclusive membership perks, dedicated managers and flexible business packages to high-frequency ones; provide seasonal bundles, flight-count rewards and scenario-based services to medium-frequency travelers; launch flash sales, entry-level membership benefits and personalized travel pushes for low-frequency users; and adopt win-back discounts, root-cause surveys and emotional connection maintenance to retain non-flyers.

6.3.4 Age- For customers of different age groups

Customers of different ages have different needs, which means that marketing strategies cannot be adopted in a one-size-fits-all manner. Instead, they should be treated differently by stage and scenario. For young customers, discounted air ticket combinations are recommended based on popular destinations on social media, such as limited-time free flights to multiple places. For middle-aged customers, family travel packages during holidays can be provided. For elderly customers, customized services such as ticket purchase discounts for the elderly, rest areas, and in-flight meals are offered.

7. Conclusion

This project has successfully established an accurate and reliable “High-Value Customer Churn Early Warning System.” It provides a data-driven customer retention strategy to enable smarter CRM (customer relationship management),

shifting the approach from proactive remediation to reactive intervention high-value customers. This transformation converts hidden churn into visible revenue.